VI SEMESTER (PROFESSIONAL ELECTIVE-II)	L	T	P	C
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20AM6E01 - HUMAN COMPUTER INTERACTION				

COURSE OBJECTIVE:

To impart knowledge on

1. To study the effectiveness of interacting with computers
2. To learn the various models that can be used for designing systems
3. To understand the importance of interaction styles
4. To learn to design web interface
5. To understand the applications of HCI

COURSE OUTCOME:

After completion of course, students would be able to:

CO1: Apply the Human Computer Interaction (HCI) principles while developing software

CO2: Identify the appropriate design model for HCI

CO3: Understand the importance of interaction styles

CO4: Postulate appropriate interaction styles Design a web interface for HCI

CO5: Apply HCI in real time systems

UNIT-I: INTRODUCTION AND DESIGN PROCESS

Interaction design basics – process of design – user focus – navigation – screen design and layout – Iteration and prototyping – HCI in the Software process – Software life cycle –Usability engineering – Iterative design and prototyping – Design Focus – Design rationale – Design Rules

UNIT -II: MODELS

Cognitive models – Goal and task hierarchies – Linguistic models – Challenge of display-based systems – Physical and device models – Cognitive architectures – Socio-organizational issues and stakeholder requirements – Socio- organizational issues and stakeholder requirements – Organizational issues – Capturing requirements – Communication and Collaboration models – Face-to-face communication – Conversation – Text-based communication – Group working.

UNIT -III: INTERACTION STYLES

Dialog notations and design –Dialog design notations – Diagrammatic notations – Textual dialog notations – Dialog semantics – Dialog analysis and design – Models of the system – Standard formalisms – Interaction models – Continuous behaviour – Modelling rich interaction – Status-event analysis – Rich contexts

UNIT -IV: WEB INTERFACE DESIGN - I

Designing Web Interfaces – Drag & Drop – Purpose – Module, List, Object, Action, Collection - Challenge – Direct Selection – Toggle, Collected, Object, Hybrid – Contextual Tools.

UNIT -V: WEB INTERFACE DESIGN - II

Overlays – Dialog, Detail, Input – Inlays – Dialog, List, Detail – Virtual Pages – Virtual Scrolling – Inline Paging – Scrolling Paging – Virtual Panning – Zoomable User Interface – Process Flow – Transitional Patterns – Purpose of Transitions.

Text Book:

1.Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, "Human Computer Interaction", 3rd edition PearsonEducation, 2009.
2.Bill Scott and Theresa Neil, "Designing Web Interfaces", First Edition, O'Reilly, 2009.

Reference Book:

1. Julie A. Jacko (Ed), "The Human-Computer Interaction Handbook", 3rd edition, CRC Press, 2012.
2. M.G. Helander, "Handbook of Human-Computer Interaction", Elsevier, 2014
3. Andrew Sears, Julie A. Jacko, "Human-Computer Interaction: Design Issues, Solutions, and Applications", CRC Press, 2009.

Extensive Reading:

1. http://nptel.ac.in/courses/106103115/
2. http://hci.bham.ac.uk/
3. www.hcibook.com/e3/online